



11. Rating Political Candidates A political scientist is studying the relationship between the voter image of a conservative political candidate and the distance (in kilometers) between the residences of the voter and the candidate. Each of 12 voters rated the candidate on a scale of 1–20.

Voter	Rating	Distance	d_u	d_v
1	12	75	2	4.5
2	7	165	1	20.5
3	5	300	12	8
4	19	15	1	12.5
5	17	180	8	4
6	12	240	11	10.5
7	9	120	4	4
8	18	60	2	8
9	3	230	10	8
10	8	200	9	16
11	15	130	5	12.5
12	4	130	5	12.5

- Calculate Spearman's rank correlation coefficient r_s .
- Do these data provide sufficient evidence to indicate a negative rank correlation between rating and distance?

11. 有關係 relationship $\Rightarrow r_s \neq 0$

$$\begin{aligned}
 a) \quad r_s &= 1 - \frac{6}{n(n^2-1)} \sum d_i^2 \\
 &= 1 - \frac{6}{12(12^2-1)} (20.25 + 9 + \dots + 12.25) \\
 &= -0.587
 \end{aligned}$$

$$b) H_0: \rho_s \geq 0$$

$$H_a: \rho_s < 0$$

$$n=12, \alpha=0.05 \Rightarrow \text{查表} = 0.497 (\text{對稱}) = CV$$

$$-0.587 < -0.497 \Rightarrow \text{Reject } H_0$$

$\Rightarrow$ 兩者負相關

12. Competitive Running Is the number of years of competitive running experience related to a runner's distance running performance? The data on nine runners, obtained from a study by Scott Powers and colleagues, are shown in the table:⁸

Runner	Years of Competitive Running	10-Kilometer Finish Time (minutes)	d_u	d_v
1	9	33.15	1	6
2	13	33.33	2	7
3	5	33.50	3	0.5
4	7	33.55	4	2
5	12	33.73	5	3
6	6	33.86	6	-1
7	4	33.90	7	-5
8	5	34.15	8	-4.5
9	3	34.90	9	-8

- Calculate the rank correlation coefficient between years of competitive running and a runner's finish time in the 10-kilometer race.
- Do the data provide evidence to indicate a significant rank correlation between the two variables? Test using $\alpha = .05$.

12.

related $\Rightarrow r_s \Rightarrow \sum d_i^2 \Rightarrow$ 兩個變量有關係
 $\Rightarrow$ 算 d

$$\begin{aligned}
 a) \quad r_s &= 1 - \frac{6}{n(n^2-1)} \sum d_i^2 \\
 &= 1 - \frac{6}{9 \times (9^2-1)} (76 + 49 + \dots + 64) \\
 &= -0.7375
 \end{aligned}$$

$$b) H_0: \rho_s = 0$$

$$H_a: \rho_s \neq 0$$

$$n=9, \alpha=0.05 \text{ 雙尾} \Rightarrow 0.683 = CV$$

$$-0.7375 < -0.683 \Rightarrow \text{Reject } H_0$$

$\Rightarrow$ 兩個變量有關係，負相關
 $\Rightarrow$ 負相關